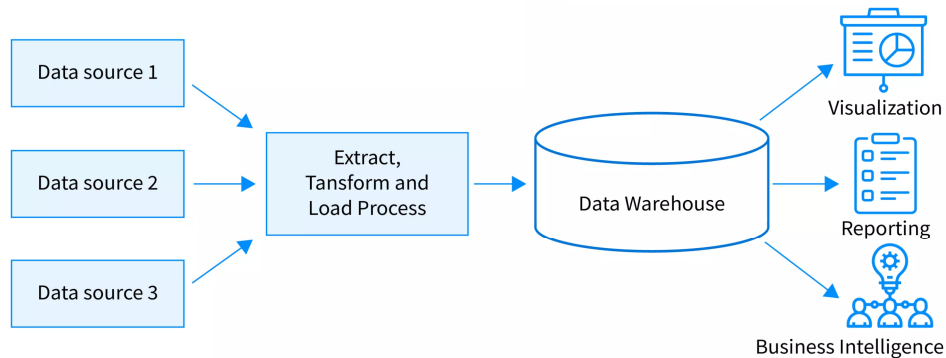


Unit-1

1. Describe the architecture of a Data Warehouse with a case study of a banking system like HDFC Bank or State Bank of India.



1. Data Source Layer

This is the first layer where data is collected from different operational systems.

Example (Banking System):

Banks like HDFC Bank or State Bank of India collect data from:

- ATM transactions
- Online banking systems
- Credit card systems
- Customer accounts
- Loan management systems

All these systems generate large amounts of daily data.

2. Data Integration Layer (ETL Process)

In this layer, the ETL process is used.

ETL stands for:

- Extract – Data is taken from different sources.
- Transform – Data is cleaned, formatted, and converted into a standard form.
- Load – Processed data is loaded into the Data Warehouse.

Example:

If a customer of HDFC Bank makes transactions through ATM, mobile banking, and net banking, the ETL process combines all this data and converts it into a single consistent format.

3. Data Warehouse Storage Layer

After processing, the data is stored in a central Data Warehouse database.

This database stores large historical data which can be used for analysis.

Example: State Bank of India stores years of customer transaction data in the warehouse to analyze:

- Customer spending patterns
- Loan repayment history
- Fraud detection patterns

4. Data Analysis and Reporting Layer

In this layer, tools are used to analyze the stored data and generate reports.

Bank managers and analysts use this information for business decisions.

Example: Using the Data Warehouse, HDFC Bank can:

- Identify customers eligible for loans
- Analyze credit card usage trends
- Detect unusual transactions for fraud prevention

Conclusion

The Data Warehouse architecture collects data from banking systems, processes it through ETL, stores it in a central database, and allows analysis for better decision-making. Banks like HDFC Bank and State Bank of India use this architecture to analyze customer data, improve services, and detect fraud.

2. How does Business Intelligence enhance decision-making in the healthcare sector? Explain with an example.

Business Intelligence (BI) in Business Intelligence helps organizations collect, analyze, and visualize data to support better and faster decision-making. In the healthcare industry, BI tools analyze patient records, hospital operations, and treatment data to improve healthcare services.

1. Better Patient Care

BI systems analyze patient data, medical history, and treatment outcomes to help doctors make better medical decisions.

Example: Hospitals like Apollo Hospitals analyze patient data to identify the best treatment plans and improve recovery rates.

Benefit:

- Accurate diagnosis
- Personalized treatment for patients

2. Disease Trend Analysis

BI tools help healthcare organizations analyze disease patterns and health trends.

Example: Organizations such as World Health Organization use data analytics to track disease outbreaks and health statistics worldwide.

Benefit:

- Early detection of diseases
- Better public health planning

3. Efficient Hospital Management

BI helps hospitals monitor staff performance, bed availability, and patient flow.

Example: Hospitals can use dashboards to check how many beds are available or how many patients are admitted daily.

Benefit:

- Better resource management
- Reduced waiting time for patients

4. Cost Reduction

Healthcare organizations use BI to analyze treatment costs, operational expenses, and resource usage.

Example: Hospitals can identify which treatments are costly and find ways to reduce unnecessary expenses.

Benefit:

- Lower healthcare costs
- Improved financial planning

5. Fraud Detection and Compliance

BI systems analyze large healthcare datasets to detect fraudulent insurance claims or unusual billing activities.

Example: Insurance companies and hospitals use BI to detect duplicate or fake medical claims.

Benefit:

- Prevents financial fraud
- Improves system transparency

6. Predictive Analysis for Healthcare

BI tools use historical data to predict future healthcare needs.

Example: Hospitals can predict future patient admissions during seasonal diseases like flu.

Benefit:

- Better preparation for emergencies

3. Explain different Data Warehouse security mechanisms with examples from financial systems.

Data warehouse security mechanisms are used to protect sensitive data from unauthorized access, theft, and misuse.

In financial systems, data warehouses store:

- Customer account details
- Transaction records
- Loan information
- Credit card data

Strong security is necessary to protect this information.

Different Security Mechanisms

1. Authentication

Meaning: Verifies the identity of users before allowing access.

Example: Bank employees log in using:

- Username
- Password
- OTP verification

2. Authorization

Meaning: Provides access only to authorized users.

Example:

- Managers can view financial reports.
- Customers cannot access internal banking data.

3. Encryption

Meaning: Converts data into coded form to prevent unauthorized access.

Example: Credit card numbers are encrypted in banking databases.

4. Access Control

Meaning: Restricts access to specific data or operations.

Types:

- Role-Based Access Control (RBAC)
- User-Level Access

Example: Only finance department employees can access loan records.

5. Data Backup and Recovery

Meaning: Creates backup copies of warehouse data to recover information during failures.

Example: Banks maintain daily transaction backups to prevent data loss.

6. Auditing and Monitoring

Meaning: Tracks user activities and system access.

Example: Bank systems record:

- Login history
- Data access logs
- Transaction updates

to detect suspicious activities.

7. Firewall and Network Security

Meaning: Protects the data warehouse from external cyber attacks.

Example: Banks use firewalls to block unauthorized network access.

Importance in Financial Systems

- Protects confidential customer information
- Prevents fraud and cyber attacks
- Ensures secure transactions
- Maintains customer trust
- Supports regulatory compliance

4. How does query optimization impact performance in a stock trading system? Explain with an example.

Query Generation in Data Warehousing refers to the process of creating and executing queries to retrieve data from a data warehouse for analysis. The way queries are written and executed can greatly affect the performance, speed, and efficiency of the data warehouse.

1. Query Complexity

When queries are very complex (many joins, filters, and calculations), they take more time to process and can slow down the system.

Example:

A stock exchange like National Stock Exchange of India analyzes millions of daily trades.

If analysts write a complex query to analyze 5 years of trading data with multiple joins, the data warehouse may take longer to generate the report.

Effect:

- Slower query response time
- Higher CPU and memory usage

2. Large Data Volume

Data warehouses store huge amounts of historical data, and queries that scan large datasets affect performance.

Example:

In financial analytics at the Bombay Stock Exchange, analysts may query years of stock price data to study market trends and trading volumes.

If the query scans the entire database, the processing time increases.

Effect:

- Long processing time
- Heavy load on the system

3. Poorly Optimized Queries

Queries that are not optimized can cause inefficient data retrieval.

Example:

A financial analyst analyzing trading patterns may write a query without proper indexes or filtering conditions.

This forces the system to search through all records, which reduces performance.

Effect:

- Slow data retrieval
- Increased server workload

4. Multiple Users Running Queries

In a data warehouse, many users run queries at the same time.

Example:

At National Stock Exchange of India, multiple analysts may simultaneously run queries for:

Stock price analysis

Trading volume reports

Risk analysis

If too many queries run together, system performance may decrease.

Effect:

- System congestion
- Slower report generation

5. Query Optimization Techniques

To improve performance, organizations use techniques such as:

- Indexing
- Query optimization
- Data partitioning
- Materialized views

Example:

Stock exchanges like Bombay Stock Exchange use optimized data warehouse systems to quickly generate reports about daily trading activity and market trends.

5. Compare centralized, distributed, and real-time Data Warehouses using examples from global e-commerce platforms.

Feature	Centralized Data Warehouse	Distributed Data Warehouse	Real-Time Data Warehouse
Definition	All data stored in a single central repository	Data distributed across multiple locations/servers	Data updated instantly as events occur
Architecture	Single server or data center	Multiple nodes across regions	Streaming + batch processing systems
Data Processing	Batch processing	Parallel processing across nodes	Stream processing (low latency)
Scalability	Limited	Highly scalable	Highly scalable
Latency	High (data delay)	Moderate	Very low (real-time insights)
Complexity	Simple to manage	Complex (network coordination needed)	Very complex (requires streaming tools)
Fault Tolerance	Low (single point of failure)	High	High

Unit-2

1. Differentiate between fact tables and dimension tables in a Star Schema.

No. Fact Table	Dimension Table
1 Stores numerical data (measures).	Stores descriptive information.
2 Contains facts like sales, profit, quantity.	Contains details like product, customer, time.
3 Used for calculation and analysis.	Used for describing and filtering data.
4 Usually very large in size.	Usually smaller in size.
5 Contains foreign keys from dimension tables.	Contains primary keys.
6 Data changes frequently with transactions.	Data changes less frequently.
7 Central table in star schema.	Surrounds the fact table in schema design.
8 Example fields: Sales_Amount, Quantity, Profit.	Example fields: Product_Name, Customer_Name, Date.
9 Stores quantitative data.	Stores qualitative data.
10 Example: Sales fact table in Flipkart storing sales amount.	Example: Product dimension table storing product details.

2. Explain the role of BI dashboards and scorecards in business performance monitoring.

1. BI Dashboards

A Business Intelligence (BI) dashboard is a visual interface that displays key data using charts, graphs, and KPIs in real time.

Role in Performance Monitoring

Real-time tracking of business activities (sales, traffic, revenue)

Data visualization makes complex data easy to understand

Helps managers quickly identify trends and issues

Supports operational decisions (day-to-day)

Example (E-commerce)

On platforms like Amazon:

Dashboard shows:

Daily sales

Active users

Order status

Managers can instantly detect a drop in sales and take action

2. Scorecards

A scorecard is a tool used to measure performance against predefined goals using KPIs (Key Performance Indicators).

Role in Performance Monitoring

- Tracks strategic goals (long-term performance)
- Compares actual vs target performance
- Helps in performance evaluation
- Aligns business activities with organizational objectives

Example (E-commerce)

On platforms like Flipkart:

Scorecard tracks:

Monthly revenue target vs achieved

Customer satisfaction score

Delivery performance

3. Explain Snowflake Schema using an example of a railway reservation system.

A Snowflake Schema is a type of data warehouse schema where:

- The fact table is connected to multiple dimension tables
- Dimension tables are further normalized into sub-dimension tables

Structure looks like a snowflake (branching shape)

Railway Reservation System Example

1. Fact Table (Central Table)

Reservation_Fact

Reservation_ID

Passenger_ID

Train_ID

Date_ID

Seat_ID

Fare

Booking_Status

Stores quantitative data (measures) like fare, bookings

2. Dimension Tables (Normalized)

Passenger Dimension

Passenger_ID

Name

Age

Gender

City_ID

➡ Further split into:

City Table → City_ID, City_Name, State

Train Dimension

Train_ID

Train_Name

Route_ID

➡ Further split into:

Route Table → Route_ID, Source_Station, Destination_Station

Date Dimension

Date_ID

Day

Month

Year

Seat Dimension

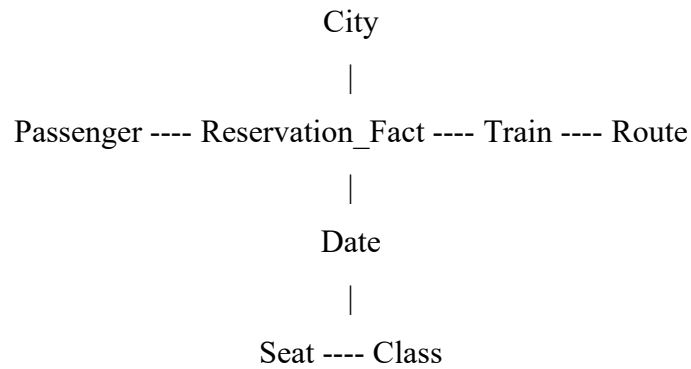
Seat_ID

Coach

Class

➡ Further split into:

Class Table → Class_ID, Class_Type (Sleeper, AC, etc.)



Role in Data Analysis

Enables detailed queries like:

- Revenue by train route

- Bookings by city

- Class-wise occupancy

- Reduces data redundancy due to normalization

4. How does granularity influence data analysis in IoT-based smart city applications?

Granularity refers to the level of detail at which data is captured and stored.

- Fine granularity → very detailed data (per second, per sensor)

- Coarse granularity → summarized data (hourly, daily averages)

Role of Granularity in Data Analysis

Accuracy of Insights

Fine granularity provides highly accurate and precise insights

Helps detect small changes and anomalies

Example: In traffic systems, second-by-second sensor data can detect congestion instantly

Data Volume & Storage

Fine granularity → huge data volume

Coarse granularity → less storage required

In smart cities (like Singapore):

- Millions of IoT sensors generate massive data continuously

Processing Speed

Fine-grained data → slower processing (heavy computation)

Coarse data → faster analysis

Real-time systems must balance speed vs detail

Decision-Making

Fine granularity → supports real-time decisions

Coarse granularity → supports strategic planning

Example:

Traffic lights adjust instantly (fine data)

City planners analyze monthly traffic patterns (coarse data)

Pattern & Trend Analysis

Coarse granularity helps identify long-term trends

Fine granularity helps identify short-term patterns

Smart City Use Cases

Traffic Management

Fine: vehicle count every second → congestion control

Coarse: daily traffic reports → infrastructure planning

Energy Management

Fine: real-time electricity usage

Coarse: monthly consumption trends

Pollution Monitoring

Fine: hourly AQI levels → health alerts

Coarse: yearly pollution trends → policy decisions

5. Explain OLAP operations (Roll-up, Drill-down, Slice, Dice) using sales analytics in Amazon.

- OLAP (Online Analytical Processing) in Online Analytical Processing is a technology used to analyze large amounts of data from different perspectives in a data warehouse.
- It helps organizations perform complex queries, data analysis, and decision-making.
- Example: E-commerce companies like Flipkart and Amazon use OLAP to analyze sales by product, region, and time.

1. Roll-up

Definition: Roll-up is the process of summarizing data by moving from detailed data to higher-level data.

Example: Flipkart analyzes daily sales → monthly sales → yearly sales.

Example hierarchy: Day → Month → Year

2. Drill-down

Definition: Drill-down is the opposite of roll-up, where data is viewed in more detailed levels.

Example: Amazon analyzes yearly sales → monthly sales → daily sales to see which day had the highest orders.

3. Slicing

Definition: Slicing means selecting a single dimension of data from a data cube.

Example: Flipkart selects sales data for only the year 2025 to analyze product sales.

Example slice: Sales of Mobile Phones in 2025.

4. Dicing

Definition: Dicing means selecting data based on multiple dimensions to create a smaller sub-cube.

Example: Amazon analyzes sales of laptops in Mumbai during January 2026.

Dimensions used:

Product = Laptop

Location = Mumbai

Time = January 2026

Unit-3

1. Explain different techniques for handling missing data using healthcare datasets (e.g., patient records).

1. Ignore or Remove Missing Data

Records with missing values are removed from the dataset.

Example: In electronic health records used by Mayo Clinic, if a patient record is missing many important fields like blood pressure and age, that record may be removed from analysis.

2. Fill with Mean, Median, or Mode

Missing values are replaced with statistical values.

Mean – average value (for numerical data)

Median – middle value

Mode – most frequent value

Example: If a patient's cholesterol value is missing, the system may fill it with the average cholesterol level calculated from other patients in the hospital database.

3. Replace with a Constant Value

A fixed value like 0, "Unknown", or "Not Available" is inserted.

Example: In electronic health records at Cleveland Clinic, if a patient's smoking status is missing, it may be filled as "Unknown."

4. Predict Missing Values Using Machine Learning

Algorithms predict the missing value based on other related attributes.

Example: If a patient's BMI value is missing, the system can estimate it using other data like weight, height, and age.

5. Interpolation Method

Used mainly for time-based medical data where values are estimated between two known values.

Example: If heart rate readings at 10:00 AM and 10:10 AM exist but the value at 10:05 AM is missing, the system estimates the value between them.

2. Summarize data transformation and discretization with reference to stock market data from National Stock Exchange of India.

1. Data Transformation

- Data Transformation is the process of converting data into a suitable format or structure for analysis in data mining. It helps improve data quality, consistency, and efficiency of data mining algorithms.
- Common Data Transformation Techniques
 - i. **Normalization**
Data values are scaled into a specific range (usually 0–1).
 - ii. **Aggregation**
Combining multiple data values into a single value.
 - iii. **Generalization**
Replacing detailed data with higher-level concepts.
 - iv. **Smoothing**
Removing noise or errors from data.
 - v. **Attribute Construction**
Creating new attributes from existing data.

Example (Case Study)

An online shopping company like Amazon collects large amounts of sales data.

Example raw data:

Customer ID Purchase Amount	
C1	5000
C2	200
C3	12000

Using normalization, the purchase values are scaled between 0 and 1 so that machine learning algorithms can process them efficiently.

2. Data Discretization

Data Discretization is the process of converting continuous data values into discrete categories or intervals. It simplifies data and makes analysis easier.

Example: Age values like 23, 25, 27, 30 can be converted into groups such as:

20–30 → Young

31–40 → Middle Age

Methods of Discretization

i. Equal Width Binning

Divides data into equal size intervals.

ii. Equal Frequency Binning

Each interval contains the same number of data points.

iii. Clustering-Based Discretization

Groups data using clustering techniques.

iv. Decision Tree Based Discretization

Uses decision tree algorithms to create intervals.

Case Study Example (Retail Industry)

An e-commerce platform like Flipkart analyzes customer purchase behavior.

Step 1: Data Transformation

Customer spending data is normalized and aggregated.

Example:

Customer	Monthly Spending
A	500
B	3000
C	12000

Step 2: Data Discretization

Spending values are converted into categories.

Spending Range	Category
0 – 1000	Low Spender
1000 – 5000	Medium Spender
Above 5000	High Spender

This helps companies understand customer segments and buying behavior.

3. Explain Outlier Analysis and its importance in fraud detection and cybersecurity applications.

- Outlier Analysis is a data mining technique used to identify data points that are very different from the normal data values. These unusual values are called outliers.
- Example: If most customers spend ₹500–₹2000, but one transaction is ₹5,00,000, it is considered an outlier.

Significance of Outlier Analysis in Data Mining

- i. Detects fraud activities in financial transactions.
- ii. Helps identify errors in data collection.
- iii. Improves accuracy of data analysis and models.
- iv. Helps detect abnormal behavior in systems.
- v. Useful in medical diagnosis to detect unusual health conditions.

Methods for Detecting Outliers

1. Statistical Method

Uses statistical measures like mean and standard deviation to find unusual values.

Example: If most patients' body temperature is 36–37°C, but one value is 40°C, it may be an outlier.

2. Distance-Based Method

A data point is considered an outlier if it is far from other data points.

Example: In a dataset of normal transactions, a transaction with extremely large value may appear far from others.

3. Density-Based Method

Compares data density around a point. If the density is very low, the point may be an outlier.

Example: Rare login attempts from an unusual location can be detected.

4. Clustering Method

Data is grouped into clusters. Points that do not belong to any cluster or are far from clusters are considered outliers.

Example: In customer segmentation, a customer with unusual purchasing behavior may appear outside clusters.

Real-World Applications

1. Fraud Detection

Banks and e-commerce companies like PayPal detect unusual transactions such as very large payments or suspicious transfers to identify fraud.

2. Cybersecurity Anomaly Detection

Companies like Microsoft analyze system logs to detect abnormal login attempts, malware activity, or unusual network traffic.

3. Medical Diagnostics

Hospitals such as Cleveland Clinic use outlier analysis to detect abnormal test results, such as extremely high blood sugar levels, which may indicate serious health problems.

4. Describe the FP-Growth algorithm with an example of market basket analysis in supermarkets like Walmart.

The FP-Growth Algorithm is a data mining technique used to find frequent itemsets in large datasets without generating many candidate sets. It is commonly used in market basket analysis to discover products that are frequently purchased together.

How FP-Growth Algorithm Works

1. Scan the Database

First, the algorithm scans the transaction database and counts the frequency of each item.

Example (Grocery store transactions):

Transaction ID	Items Purchased
T1	Milk, Bread
T2	Milk, Bread, Butter
T3	Bread, Butter
T4	Milk, Bread
T5	Milk, Butter

Frequency count:

Milk → 4

Bread → 4

Butter → 3

2. Remove Infrequent Items

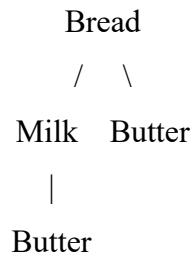
Items with frequency below the minimum support threshold are removed.

Example: If minimum support = 2, all items remain because they appear more than twice.

3. Build FP-Tree

The algorithm builds an FP-tree (Frequent Pattern Tree) structure that stores items in a compressed form.

Example structure:



This tree represents frequent item combinations.

4. Generate Frequent Patterns

The algorithm extracts frequent itemsets from the FP-tree.

Example results:

Milk → Bread

Bread → Butter

Milk → Butter

These patterns show items that are commonly bought together.

Real-World Example (Market Basket Analysis)

A supermarket like Walmart uses FP-Growth to analyze customer shopping baskets.

Example finding:

- Customers who buy Bread often buy Butter.
- Customers who buy Milk often buy Bread.

Business benefits:

- Place related products close together in the store.
- Create combo offers like “Buy Bread and Butter together”.
- Improve sales and marketing strategies.

5. Explain Decision Trees (ID3, C4.5, CART) and their applications in spam filtering and medical diagnosis.

- A Decision Tree is a supervised machine learning technique used for classification and prediction.
- It represents decisions and their possible outcomes in a tree-like structure.
- Each branch represents a decision rule, and each leaf node represents a final result or class.
- Example: An email system like Gmail uses decision trees to classify emails as Spam or Not Spam based on words, sender information, and links.

1. Structure of a Decision Tree

A decision tree has three main components.

1. Root Node

The top node of the tree.

Represents the main attribute used for decision making.

Example: In medical diagnosis, the root node might be “Does the patient have fever?”

2. Internal Nodes

These nodes represent conditions or tests on attributes.

Each node splits the data based on a condition.

Example: If Fever = Yes, check Cough = Yes/No.

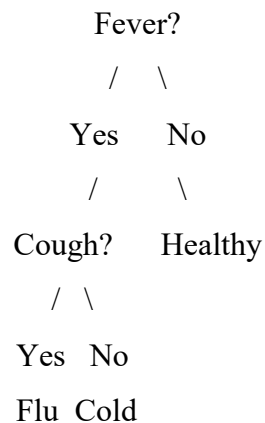
3. Leaf Node

The final node of the tree.

Shows the final decision or classification result.

Example:

Disease = Flu.



2. Working Mechanism of Decision Trees

Decision trees work by splitting the dataset into smaller subsets based on the best attribute.

Step 1: Select Best Attribute

The algorithm chooses the attribute that best separates the data.

Example: Attributes like age, income, purchase history in customer analysis.

Step 2: Split the Data

The dataset is divided into branches based on attribute values.

Example: Income → High / Medium / Low.

Step 3: Create New Nodes

For each branch, the algorithm again selects the best attribute and splits the data.

Step 4: Continue Until Stop Condition

The process stops when:

- All data belongs to the same class
- No more attributes remain

3. Types of Decision Tree Algorithms

1. ID3 (Iterative Dichotomiser 3)

The ID3 algorithm selects attributes using Information Gain.

Features

- Works mainly with categorical data
- Simple and easy to implement

Example: Used in email spam classification.

Limitation: Cannot handle continuous values well.

2. C4.5 Algorithm

The C4.5 algorithm is an improved version of ID3.

Features

- Uses Gain Ratio instead of information gain
- Handles continuous data
- Handles missing values

Example: Used in customer behavior analysis.

3. CART (Classification and Regression Tree)

The CART algorithm creates binary trees (two branches).

Features

- Works for classification and regression
- Uses Gini Index to split data

Example: Used in financial risk prediction and medical diagnosis.

4. Advantages of Decision Trees

1. Easy to Understand

Decision trees are simple and can be visualized easily.

2. Requires Less Data Preparation

They work even if the dataset has missing values or non-linear relationships.

3. Handles Different Types of Data

Decision trees can handle both:

- Numerical data
- Categorical data

4. Useful for Decision Making

Managers can easily interpret the decision rules.

5. Fast Prediction

Once the tree is built, predictions are very fast.

5. Disadvantages of Decision Trees

1. Overfitting Problem

Decision trees may fit the training data too closely, reducing accuracy on new data.

2. Instability

Small changes in data may create a completely different tree.

3. Complex Trees

Large datasets may produce very large trees, which are difficult to interpret.

4. Bias Toward Dominant Attributes

Some attributes may dominate the decision process.

6. Real-World Applications

1. Spam Detection

Email services like Gmail use decision trees to detect spam emails.

The system checks:

- Sender address
- Keywords
- Suspicious links

If conditions match spam patterns, the email is classified as Spam.

2. Medical Diagnosis

Hospitals use decision trees to detect diseases.

Example conditions:

- Fever
- Blood pressure
- Blood sugar level

Hospitals such as Mayo Clinic use predictive models to assist doctors in diagnosing diseases.

3. Customer Segmentation

E-commerce companies like Amazon analyze customer behavior using decision trees.

Example:

Customers are grouped based on:

- Purchase frequency
- Spending amount
- Product preferences

This helps companies create personalized recommendations and marketing strategies.